

Customer Churn and Revenue Analysis Dashboard

Project Title

Customer Churn and Revenue Analysis Using Power BI

Project Overview

This project analyzes customer behavior, revenue generation, customer loyalty, device usage, and churn patterns using an e-commerce customer dataset. The dashboard was developed in Power BI to help business stakeholders understand customer retention trends and identify opportunities to improve revenue and customer engagement.

Business Problem

Customer retention is critical for long-term business growth. The objective of this project was to identify factors affecting customer churn, evaluate customer spending behavior, and provide data-driven recommendations for improving customer retention and revenue generation.

Tools Used

- Power BI
 - DAX
 - Data Modeling
 - Data Visualization
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Key Performance Indicators (KPIs)

- Total Customers: 1,000
 - Total Orders: 65,000
 - Total Revenue: \$6.36 Million
 - Average Return Rate: 166.28
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Dashboard Components

Customer Churn by Loyalty Tier

The analysis compares churn levels across Gold, Silver, Platinum, and Bronze loyalty tiers.

Revenue by Country

Revenue contribution was analyzed across multiple countries to identify top-performing markets.

Revenue by Preferred Category

Customer spending was evaluated across product categories to identify the most profitable segments.

Revenue by Referral Source

The dashboard measures the effectiveness of acquisition channels by comparing total revenue generated from each source.

Device Usage Analysis

Customer distribution across Mobile, Desktop, and Tablet devices was analyzed to understand platform preferences.

Churn by Device Type

Customer churn was compared across device types to identify potential differences in retention behavior.

Customer Spending by Age Group

Revenue contribution by age group was examined to identify the most valuable customer demographics.

Key Findings

1. Strong Revenue Performance

The business generated approximately \$6.36 million in total revenue from 1,000 customers, indicating a high customer value base.

2. Mobile Devices Dominate Usage

More than half of customers use mobile devices, making mobile the most important platform for customer engagement.

3. Loyalty Programs Influence Customer Behavior

Gold, Silver, and Platinum loyalty tiers show strong customer participation, indicating that loyalty programs play an important role in customer retention.

4. Revenue Concentration in Key Markets

A small number of countries contribute a significant share of overall revenue, suggesting opportunities for focused regional marketing strategies.

5. Referral Sources Impact Revenue Generation

Certain referral channels generate substantially more revenue than others, indicating that customer acquisition quality varies across marketing channels.

6. Product Preferences Affect Revenue

Several product categories consistently generate higher spending, demonstrating the importance of product mix optimization.

7. Age Influences Customer Spending

Middle-age customer groups contribute the highest revenue, making them a valuable target audience for marketing campaigns.

8. Device Type May Influence Churn

Customer churn patterns vary across device types, suggesting opportunities to improve user experience on specific platforms.

Business Recommendations

Recommendation 1

Increase investment in loyalty programs to encourage repeat purchases and improve customer retention.

Recommendation 2

Prioritize mobile user experience optimization because the majority of customers access services through mobile devices.

Recommendation 3

Allocate more marketing resources toward the highest-performing referral channels to maximize return on investment.

Recommendation 4

Develop targeted campaigns for high-value age groups that generate the largest revenue contribution.

Recommendation 5

Investigate countries with lower revenue performance and explore localized promotional strategies.

Recommendation 6

Focus inventory planning and marketing efforts on top-performing product categories.

Recommendation 7

Implement customer retention campaigns targeting users at higher risk of churn.

Conclusion

The Power BI dashboard successfully provides a comprehensive overview of customer behavior, revenue performance, loyalty participation, and churn trends. The analysis highlights the importance of customer retention strategies, loyalty programs, mobile optimization, and targeted marketing initiatives. These insights can support data-driven decision-making and help improve long-term business performance.

Skills Demonstrated

- Data Cleaning
- Data Analysis
- Business Intelligence
- Dashboard Development
- Data Visualization
- KPI Development
- DAX Measures
- Customer Churn Analysis
- Revenue Analysis
- Business Recommendations